

On the Fixed-Order Strong Komlós Conjecture

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Abstract

For an ordered family $v_1, \dots, v_n \in \mathbb{R}^d$ with $\|v_i\|_2 \leq 1$, its fixed-order prefix discrepancy is the least, over all signings, of the largest ℓ_∞ -norm of a signed prefix. We construct explicit finite families for which this quantity is unbounded. At level k , the raw matrix has

$$d_k = 2^{2^k - 1} \text{ rows and } s_k = 2^k$$

entries from $\{-1, 1\}$ in every column. For every real coefficient assignment whose entries have magnitude at least one, some coordinate of the partial-sum path has range at least s_k . After Euclidean normalization this gives prefix discrepancy at least

$$\frac{1}{2} \sqrt{s_k} = \frac{1}{2} \sqrt{1 + \log_2 d_k} \rightarrow \infty.$$

Our proof uses a finite nested word that freezes one large mean at each depth and then applies a detector matching the selected row tuple. We also obtain square $N \times N$ examples with lower bound $\Omega(\sqrt{\log \log N})$ and closed examples with zero final signed sum.

1 Introduction

For an ordered real matrix $A = (v_1, \dots, v_n)$, define

$$\text{pdisc}(A) = \min_{\varepsilon \in \{-1, 1\}^n} \max_{0 \leq j \leq n} \left\| \sum_{i=1}^j \varepsilon_i v_i \right\|_\infty. \quad (1.1)$$

The *strong*, or fixed-order prefix, Komlós assertion asks whether there is an absolute constant C such that $\text{pdisc}(A) \leq C$ whenever $\|v_i\|_2 \leq 1$, independently of d and n . The signs in (1.1) may depend on the entire ordered matrix; this is an offline problem, and the input order cannot be changed.

Ordinary Komlós discrepancy controls only the final signed sum. It was recently bounded by $3\sqrt{2\pi}$ by Guo, Fang, and Lu [4], and Karingula and Lovett subsequently gave an elementary proof with constant 36 [5]. Kunisky constructed ordinary lower examples approaching $1 + \sqrt{2}$ [6]. These endpoint results allow large intermediate excursions and therefore do not imply a prefix bound.

For fixed order, Bansal, Jiang, Meka, Singla, and Sinha record the general estimate $\text{pdisc}(A) = O(\sqrt{\log d + \log n})$ and pose the universal-constant problem explicitly [3]. They also improve the dependence on n after random smoothing. Aden-Ali gives an online $O(\sqrt{\log n})$ bound for a sequence fixed independently of the algorithm’s randomness [1]. Altschuler and

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Tikhomirov obtain the $O(\sqrt{s})$ prefix scale for sparse binary columns when the sparsity is at least $(\log n)^{1+o(1)}$ [2]. In fixed dimension the behavior is different: Lund and Magazinov prove that the Euclidean-plane sign-sequence constant is exactly $\sqrt{3}$ [7].

Our main theorem gives an unbounded lower bound in growing dimension. Our construction is robust: the same range conclusion holds when each sign is replaced by an arbitrary real coefficient of magnitude at least one. Its length grows extremely rapidly, so the result is qualitative rather than efficient. For square matrices our present lower bound is $\Omega(\sqrt{\log \log N})$, leaving a gap to the known $O(\sqrt{\log N})$ upper bound.

2 The robust amplification construction

2.1 Centered and robust prefix discrepancy

For $c = (c_1, \dots, c_n) \in \mathbb{R}^n$, write

$$S_j(A, c) = \sum_{i=1}^j c_i v_i, \quad S_0(A, c) = 0.$$

For a nonempty coefficient set $E \subseteq \mathbb{R}$, let

$$\text{cdisc}_E(A) = \inf_{\substack{c \in E^n \\ w \in \mathbb{R}^d}} \max_{0 \leq j \leq n} \|w + S_j(A, c)\| * \infty. \quad (2.1)$$

When $E = \{-1, 1\}$, we write simply $\text{cdisc}(A)$ and call this the centered prefix discrepancy. Because the path is finite,

$$\inf_{w \in \mathbb{R}^d} \max_{0 \leq j \leq n} \|w + S_j(A, c)\| * \infty = \frac{1}{2} \max_{1 \leq r \leq d} \left(\max_j [S_j(A, c)]_r - \min_j [S_j(A, c)]_r \right). \quad (2.2)$$

Indeed, each coordinate is optimally translated to the midpoint of its minimum and maximum. In particular, $\text{cdisc}(A) \leq \text{pdisc}(A)$. It is convenient to use the following pointwise form of (2.1).

Definition 2.1. For $a \geq 0$, let $P_E(A, a)$ denote the statement

$$\text{for every } c \in E^n \text{ and } w \in \mathbb{R}^d, \quad \max_{0 \leq j \leq n} \|w + S_j(A, c)\|_\infty \geq a. \quad (2.3)$$

Thus $P_E(A, a)$ is equivalent to $\text{cdisc}_E(A) \geq a$, or, by (2.2), to a maximum-coordinate range of at least $2a$ for every permitted coefficient assignment. The arbitrary offset w is the key feature that allows sequential copies of the construction to reuse coordinates.

2.2 The nested word

Let $M = (m_{rj})$ be a $d \times n$ ordered matrix, with $d, n \geq 1$. The output coordinates are indexed by triples

$$(\ell, r, \sigma), \quad \ell, r \in [d], \quad \sigma \in \{+, -\}.$$

There are $2d^2$ coordinates. Let $e_{\ell, r, \sigma}$ be the corresponding coordinate vector.

For every depth $\ell \in [d]$ and source column $j \in [n]$, define an *original* column

$$O_{\ell, j} = \sum_{r=1}^d m_{rj} (e_{\ell, r, +} + e_{\ell, r, -}). \quad (2.4)$$

For every tuple $\rho = (r_1, \dots, r_d) \in [d]^d$, define a detector word $D_{\rho,1}, \dots, D_{\rho,n}$ by

$$D_{\rho,j} = \sum_{\ell=1}^d m_{\ell j} (e_{\ell,r_\ell,+} - e_{\ell,r_\ell,-}). \quad (2.5)$$

In (2.5), the source row is ℓ ; the entry r_ℓ chooses only the physical coordinate pair into which that source row is placed. Different depths always use disjoint pairs.

Let Q be the concatenation, in lexicographic order, of the d^d detector words. Define words $W_{d+1}, W_d, \dots, W_1$ backwards by

$$\begin{aligned} W_{d+1} &= Q, \\ W_\ell &= W_{\ell+1}, O_{\ell,1}, W_{\ell+1}, O_{\ell,2}, \dots, O_{\ell,n}, W_{\ell+1} \quad (\ell = d, \dots, 1). \end{aligned} \quad (2.6)$$

The output $F(M)$ is the matrix whose ordered column word is W_1 . Every physical occurrence in (2.6) receives its own coefficient: repeated column vectors do not share signs or coefficients. The word is completely fixed before those coefficients are chosen.

Every original and detector column corresponding to M_j satisfies

$$\|O_{\ell,j}\| * 2^2 = \|D * \rho, j\|_2^2 = 2 \|M_j\|_2^2. \quad (2.7)$$

There are no coordinate collisions in either sum. In particular, if every source column has exactly s nonzero entries, all equal to ± 1 , then every output column has exactly $2s$ such entries.

If n' and d' are the length and dimension of $F(M)$, then

$$d' = 2d^2, \quad n' = (1 + nd^d)(n+1)^d - 1. \quad (2.8)$$

Indeed, $|W_{d+1}| = nd^d$ and $|W_\ell| = (n+1)|W_{\ell+1}| + n$.

2.3 The amplification lemma

Lemma 2.2 (Nested mean-difference amplification). *If $P_E(M, a)$ holds, then $P_E(F(M), 2a)$ holds.*

Proof. Fix arbitrary coefficients from E on all physical occurrences in $F(M)$, and fix an arbitrary initial offset in $\mathbb{R}^{2d^2}$. For each pair of physical coordinates write

$$p_{\ell,r,+} = h_{\ell,r} + z_{\ell,r}, \quad p_{\ell,r,-} = h_{\ell,r} - z_{\ell,r}. \quad (2.9)$$

An original column at depth ℓ changes the mean vector $h_{\ell,\cdot}$ by the corresponding source column multiplied by its actual coefficient. It changes no tracker and no mean at another depth. Every detector preserves all means and changes only trackers.

Consider one occurrence of W_ℓ . The only columns in that occurrence that change $h_{\ell,\cdot}$ are $O_{\ell,1}, \dots, O_{\ell,n}$, in that order. Therefore, at the starts of its $n+1$ displayed child occurrences of $W_{\ell+1}$, the depth- ℓ means form

$$h_0, \quad h_0 + c_1 M_1, \quad \dots, \quad h_0 + \sum_{j=1}^n c_j M_j, \quad (2.10)$$

where the $c_j \in E$ are the coefficients of these particular original occurrences. By $P_E(M, a)$, some row r_ℓ and some slot in (2.10) satisfy

$$|h_{\ell,r_\ell}| \geq a. \quad (2.11)$$

No column in the corresponding child $W_{\ell+1}$ changes this mean.

Begin with W_1 , select a child slot as above, and repeat inside that selected child. Induction on $\ell = 1, \dots, d$ produces one actual occurrence of Q and rows $r_1, \dots, r_d$ such that throughout that entire occurrence

$$|h_{\ell, r_\ell}| \geq a \quad (\ell \in [d]). \quad (2.12)$$

All these means are simultaneously constant. The selection is a post hoc choice of a witness interval in the already fixed word; it does not alter the word or its coefficients.

The selected Q contains the detector indexed by $\rho = (r_1, \dots, r_d)$. During that detector, (2.5) shows that $(z_{1, r_1}, \dots, z_{d, r_d})$ is an arbitrary translate of an M -path with coefficients in E . Earlier detectors may have changed its initial value, which is harmless because P_E permits every offset. Hence, at some prefix of this detector, one selected tracker satisfies

$$|z_{\ell, r_\ell}| \geq a. \quad (2.13)$$

At the same global time, (2.12) still holds. The scalar identity $\max\{|h+z|, |h-z|\} = |h| + |z|$ then gives a physical coordinate of magnitude at least $2a$. The coefficient assignment and initial offset were arbitrary, which is exactly $P_E(F(M), 2a)$. $\square$

Remark 2.3. The proof does not require a detector to start or end at zero. Copies at the same depth may leave arbitrary offsets for later copies because (2.3) is offset-robust. Ancestor means remain frozen inside the selected descendant interval, and different depths use disjoint coordinate pairs.

3 Counterexamples

3.1 The main counterfamily

Theorem 3.1. *For every integer $k \geq 0$, there is an explicitly specified finite ordered integer matrix M_k with*

$$d_k = 2^{2^k-1} \text{ rows,} \quad s_k = 2^k$$

nonzero entries in every column, all belonging to $\{-1, 1\}$, such that every real coefficient assignment $|c_i| \geq 1$ has a coordinate whose partial-sum trajectory, including prefix zero, has range at least s_k . Consequently, $A_k = M_k / \sqrt{s_k}$ has Euclidean-unit columns and

$$\text{pdisc}(A_k) \geq \frac{\sqrt{s_k}}{2} = 2^{k/2-1} = \frac{1}{2} \sqrt{1 + \log_2 d_k} \longrightarrow \infty. \quad (3.1)$$

Proof. Let $E = \{c \in \mathbb{R} : |c| \geq 1\}$, take $M_0 = [1]$, and define $M_{k+1} = F(M_k)$. For $c \in E$,

$$\max\{|w|, |w+c|\} \geq |c|/2 \geq 1/2,$$

so $P_E(M_0, 1/2)$ holds. Lemma 2.2 gives

$$P_E(M_k, 2^{k-1}). \quad (3.2)$$

By (2.2), every permitted path therefore has maximum coordinate range at least 2^k .

Equations (2.7) and (2.8) give

$$s_{k+1} = 2s_k, \quad d_{k+1} = 2d_k^2, \quad s_0 = d_0 = 1.$$

Thus $s_k = 2^k$ and $d_k = 2^{2^k-1}$. Every recursively specified length is finite. Dividing by $\sqrt{s_k}$ makes every column have Euclidean norm exactly one. Applying (3.2) with zero offset and coefficients $\varepsilon_i \in \{-1, 1\}$ proves (3.1). Since the lower bound is positive, its witness cannot be the empty prefix. $\square$

3.2 Square instances

Corollary 3.2 (Square matrices). *For every integer $N \geq 1$, there is an $N \times N$ real matrix B_N with Euclidean-unit columns such that*

$$\text{pdisc}(B_N) \geq \frac{1}{2\sqrt{2}} \sqrt{1 + \log_2 \log_2(N+1)}. \quad (3.3)$$

Proof. Let n_k be the number of columns of M_k . We claim

$$n_k + 1 \leq 2^{d_k}. \quad (3.4)$$

This is equality for $k = 0$. If $n + 1 \leq 2^d$, then (2.8) gives

$$\begin{aligned} n' + 1 &= (1 + nd^d)(n + 1)^d \\ &\leq d^d (n + 1)^{d+1} \\ &\leq 2^{d \log_2 d + d(d+1)} \leq 2^{2d^2} = 2^{d'}, \end{aligned}$$

where the last inequality is equivalent to $\log_2 d + 1 \leq d$. Also, if $n \geq d$, then

$$n' \geq nd(n + 1) \geq 2d^2 = d',$$

so $d_k \leq n_k$ by induction. The lengths increase without bound.

Choose k with $n_k \leq N < n_{k+1}$. By (3.4),

$$1 + \log_2 \log_2(N + 1) \leq 1 + \log_2 d_{k+1} = 2s_k. \quad (3.5)$$

Pad A_k with zero rows to dimension N , then append $N - n_k$ copies of a unit coordinate column. Every signing of the resulting $N \times N$ matrix restricts to a signing of its initial A_k block. Combining (3.1) and (3.5) proves (3.3). $\square$

4 Endpoint behavior and model separation

4.1 Closed counterexamples

Proposition 4.1. *Let $E = \{c \in \mathbb{R} : |c| \geq 1\}$, let $C_0 = [1, 1]$, and define $C_{k+1} = F(C_k)/\sqrt{2}$. Every column of C_k has Euclidean norm one, and*

$$\text{cdisc}_E(C_k) = \text{cdisc}(C_k) = 2^{k/2-1}. \quad (4.1)$$

Moreover, the upper bound is attained by a signing whose total signed sum is zero.

Proof. For C_0 , the signing $(+1, -1)$, centered at $-1/2$, stays within radius $1/2$ and returns to zero. Also $P_E(C_0, 1/2)$ holds because the first step has magnitude at least one.

Suppose the assertion holds for a source family C , with centered radius q , a closing signing σ , and a center b . Lemma 2.2, followed by division by $\sqrt{2}$, gives the robust lower bound $\sqrt{2}q$ for $F(C)/\sqrt{2}$. For the matching upper bound, use σ in every original and detector copy and initialize the pair means and trackers by

$$h_{\ell,r}(0) = b_r/\sqrt{2}, \quad z_{\ell,r}(0) = b_\ell/\sqrt{2}.$$

Every completed copy returns to this baseline. During a copy, each active mean or tracker follows a coordinate of $(b + S_j(C, \sigma))/\sqrt{2}$, while all others remain at baseline. Hence $|h_{\ell,r}|, |z_{\ell,r}| \leq q/\sqrt{2}$, and therefore $|h_{\ell,r} \pm z_{\ell,r}| \leq \sqrt{2}q$. The constructed signing closes because every constituent copy closes. It belongs to E , so it supplies the upper bound for both cdisc_E and cdisc . Finally, (2.7) shows that division by $\sqrt{2}$ preserves unit column norms. Induction gives (4.1). $\square$

4.2 Final sums and free reordering

For an ordered matrix A , let

$$\text{disc}(A) = \min_{\varepsilon \in \{-1,1\}^n} \left\| \sum_{i=1}^n \varepsilon_i A_i \right\|_{\infty}$$

denote its ordinary final-sum discrepancy.

Proposition 4.2. *Let n_k be the number of columns of the basic family $A_k = M_k/\sqrt{s_k}$. Then*

$$\text{disc}(A_k) = \frac{1}{\sqrt{s_k}}, \quad (4.2)$$

and its freely reordered prefix discrepancy is

$$\min_{\substack{\pi \in \mathfrak{S}_{n_k} \\ \varepsilon \in \{-1,1\}^{n_k}}} \max_{0 \leq j \leq n_k} \left\| \sum_{i=1}^j \varepsilon_i(A_k) * \pi(i) \right\| * \infty = \frac{1}{\sqrt{s_k}}. \quad (4.3)$$

Proof. We prove two facts about the raw matrix M_k . Let $j_k = (n_k + 1)/2$. First, there is a probability row vector f_k , supported on exactly s_k rows and equal to $1/s_k$ on that support, such that

$$f_k^T M_k = e_{j_k}^T. \quad (4.4)$$

For $M_0 = [1]$, take $f_0 = (1)$. If $f^T M = e_j^T$, define g on the output coordinates by

$$g_{1,r,+} = g_{1,r,-} = f_r/2, \quad g_{\ell,r,\sigma} = 0 \quad (\ell > 1).$$

Every detector cancels against g , and every original below the first depth is disjoint from its support. For a first-depth original, $g^T O_{1,i} = f^T M_i$. Thus g isolates the occurrence of the source's distinguished column. If their common length is L , that occurrence is at position

$$j(L+1) = \frac{(n+1)(L+1)}{2} = \frac{n'+1}{2},$$

so it is the central output column. This proves (4.4) by induction.

Second, the central column-vector type is the unique type of odd multiplicity. Equation (2.8) preserves odd length, because $n+1$ is even whenever n is odd; hence every n_k is odd. In $F(M)$, an original copy at depth ℓ occurs $(n+1)^{\ell-1}$ times, while each tuple-detector copy occurs $(n+1)^d$ times. Modulo two, all contributions vanish except the single first-depth original embedding. That embedding is injective on source column types, so it preserves the unique odd type.

For every signing ε , (4.4) and the fact that f_k is a probability vector give

$$\|M_k \varepsilon\| * \infty \geq |f_k^T M_k \varepsilon| = |\varepsilon * j_k| = 1. \quad (4.5)$$

Conversely, pair equal-column occurrences and give opposite signs to each pair, leaving one occurrence of the unique odd type unpaired. The final sum is that one raw column and has ℓ_{∞} -norm one. This proves (4.2) after normalization.

For (4.3), place each equal pair consecutively with opposite signs and put the unpaired occurrence last. Every raw prefix is then either zero or one signed column, so its norm is at most one. Conversely, under any permutation and signing the first prefix has norm one. Division by $\sqrt{s_k}$ proves (4.3). $\square$

Use of AI: The author wrote a draft with an outline of the construction (split each coordinate into a mean and tracker, use nesting to freeze large means, try every possible row tuple and iterate) and fed this draft to ChatGPT and prompted it to work out the details, identify missing gaps and disprove Strong Komlós conjecture. The author fixed those missing gaps and iteratively verified and expanded the proofs with ChatGPT. The author is solely responsible for the correctness of the proofs.

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